Removing AI Watermarks with Off-the-Shelf Diffusion Models: A Practical Demonstration

Diffusion Watermarking Project

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1 What Is a Diffusion Model?

A diffusion model is a neural network that learns to generate images by practising a very specific skill: starting from pure random noise and gradually turning it into a coherent picture, one small step at a time. Training works in reverse — the model is shown real images with increasing amounts of noise added, and learns to predict how to remove the noise at each step. Once trained, you can hand it pure static and it will denoise its way to a photorealistic image.

The key capability we exploit is called **image-to-image** (img2img) regeneration. Instead of starting from pure noise, you start from an existing image, add a controlled amount of noise (destroying some of the original signal), and then let the model denoise it back into a clean image. The result looks similar to the original — same composition, colours, and subject — but every pixel has been *regenerated* by the diffusion model. The "strength" parameter controls how much noise is added: low strength preserves more of the original; high strength gives the model more creative freedom.

This is not a research prototype. Stable Diffusion is free, open-source software that runs on a consumer laptop. The img2img pipeline is a single function call in the standard library.

2 The Attack

Digital watermarks work by hiding a signal in the pixel values of an image. The signal is imperceptible to the human eye but detectable by the right algorithm. The watermark's security assumption is that an attacker cannot remove this signal without visibly degrading the image.

The img2img attack breaks this assumption. When a watermarked image is passed through the diffusion model:

1. **Noise destroys the watermark.** The forward diffusion step adds Gaussian noise to the image, directly corrupting the learned signal pattern that the watermark encoder embedded.

2. **Denoising reconstructs the image, not the watermark.** The model reconstructs a visually similar image from the noisy version, but the reconstruction is guided by the diffusion model's own learned prior — not by the watermark encoder. The output carries no trace of the original watermark.

No training or fine-tuning is required. The attacker does not need to know anything about the watermark's architecture, key, or detection method. A single forward pass through an unmodified, off-the-shelf Stable Diffusion 1.5 checkpoint is sufficient.

3 Results: VideoSeal (Meta)

We first tested the attack against **VideoSeal**, Meta's open-source neural watermark. VideoSeal embeds a 256-bit message into images using a learned neural encoder/decoder — it is the closest publicly available analogue to Google's SynthID. We embedded watermarks at an above-default strength (`scaling_w` = 0.3, 50% above the default) into 5 images from DiffusionDB, then attacked each at 3 strength levels with 2 guidance scales (6 combinations per image, 30 runs total).

Table 1: VideoSeal attack results, averaged over 5 images. Bit accuracy at chance level (50%) means no watermark signal remains.

Attack Strength	SSIM	PSNR (dB)	Bit Accuracy	Watermark Detected
Before attack	*0.991*	*41.6*	*100.0%*	*5/5*
0.3 (mild)	0.540	20.4	53.1%	0/5
0.5 (moderate)	0.433	18.2	49.9%	0/5
0.7 (strong)	0.337	16.0	49.6%	0/5
Chance level	—	—	*50.0%*	—

The watermark was removed in 100% of trials (30/30), at every attack strength tested. Even the mildest setting (strength 0.3) reduced bit accuracy from 100% to 53% — statistically indistinguishable from random guessing. The p-value collapsed from 10^{-78} (before attack) to $\sim$0.1 (after), a drop of 77 orders of magnitude.

The attacked images are visibly altered (SSIM drops to 0.34–0.54) but remain recognisable. At strength 0.3, image quality is roughly comparable to moderate JPEG compression.

4 Pilot Test: SynthID (Google)

Having established the attack on VideoSeal, we turned to the real target: **SynthID**, Google DeepMind's 136-bit watermark deployed across all Gemini-generated images. SynthID's detector is closed-source, but Google provides a public detection portal that returns a binary yes/no verdict.

For the pilot, we generated a single image via the Gemini API ("old man in a boat", 1408×768). Because Stable Diffusion operates at 512×512, we used a **tiled attack**: split the image into overlapping patches, attack each tile independently, then reassemble. We tested three strengths and uploaded each result to Google's detection portal.

Table 2: Pilot SynthID attack ($n = 1$). Detection via Google's SynthID portal.

Attack Strength	SSIM	PSNR (dB)	SynthID Detected
Original image	*1.000*	∞	*Yes*
0.2	0.684	22.1	**No**
0.3	0.617	20.7	**No**
0.5	0.472	18.3	**No**

All three attack strengths removed the SynthID watermark according to Google's own detector. This is only $n = 1$, but it demonstrates that the attack concept transfers from VideoSeal to SynthID without any modification.

5 Scaling Up: 50-Image SynthID Corpus

To move beyond a single data point, we generated a **50-image corpus** via the Gemini API using diverse prompts (landscapes, portraits, abstract art, food, sci-fi, etc.; all 1024×1024 PNG). Each image is attacked at six fine-grained strengths using the same tiled img2img pipeline.

Since we cannot submit 300 images to Google's portal manually, we use **reverse-SynthID**, an open-source detector that analyses Fourier-domain carrier frequencies characteristic of SynthID. While less authoritative than Google's detector, it provides continuous metrics (correlation strength, phase coherence) rather than just a binary verdict, letting us track exactly how the watermark signal degrades.

Table 3: 50-image SynthID corpus sweep. Results pending — sweep in progress at time of writing.

Attack Strength	Mean SSIM	Mean PSNR	Mean Correlation	Detection Rate
0.05	—	—	—	—/50
0.10	—	—	—	—/50
0.15	—	—	—	—/50
0.20	—	—	—	—/50
0.25	—	—	—	—/50
0.30	—	—	—	—/50
Original	*1.000*	∞	*TBD*	*50/50*

Preliminary results on 2 images show that correlation drops by $\sim 3\times$ between the original and strength 0.3, confirming the watermark signal is being degraded. The full sweep ($50 \times 6 = 300$ attacks) will reveal whether SynthID has a sharp breaking point or a gradual decline.

6 What This Means

The core finding is simple: **an off-the-shelf diffusion model, used as-is, removes state-of-the-art neural watermarks.**

- **No expertise required.** The attack is a single function call in a free, open-source library. Anyone who can install Python can run it.
- **No knowledge of the watermark needed.** The attacker does not need to know the watermark's architecture, key, embedding strength, or detection algorithm. The diffusion model is not "trained against" the watermark — it simply destroys the signal as a side effect of its normal operation.
- **Transfers across watermark systems.** The same attack (same model, same code, same parameters) removed both VideoSeal (Meta, 256-bit) and SynthID (Google, 136-bit) watermarks, despite having no access to either system's internals.
- **Quality–robustness tradeoff favours the attacker.** Even the mildest attack setting (strength 0.3) completely removes the watermark while preserving recognisable image content. The attacker can choose how much quality to sacrifice.

This does not mean watermarking is useless — it raises the bar for casual redistribution and provides a signal for platforms that control the upload pipeline. But it does mean that any motivated individual can strip the watermark in seconds, which limits the technology's value as a forensic provenance tool against adversarial actors.

The watermark cannot survive regeneration because it is embedded in pixel-level statistics, and the diffusion model regenerates those statistics from scratch. Any watermark that operates in the same signal domain as the diffusion model's output is fundamentally vulnerable to this class of attack.